Yamin Haque

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PROFESSIONAL SUMMARY

Software Engineer (R&D) with around three years developing and debugging C / Embedded C software for ARM-based carrier Ethernet switches running embedded Linux. Experienced in Layer 2 networking, Carrier Ethernet protocols, routing and switching, network security, and standards-based protocol implementation. Strong expertise in low-level C programming, including pointers, bitwise operations, type casting, structure packing, byte-order handling, and memory management. Skilled in HAL/driver integration, event-driven hardware notification handling, TCP/UDP communication, kernel tracing, performance analysis, and embedded debugging. Comfortable with command-line, no-IDE development and debugging workflows, with a solid foundation in microcontrollers and RTOS. Familiar with AI and deep learning technologies, and experienced in collaborating with overseas engineering teams and clients using professional working English.

WORK EXPERIENCE

Software Engineer (R&D) 2023 Apr - 2026 July
Shanghai BDCOM Information Technology Co., Ltd. (Bangladesh office)
Rangs Paramount-2, Level-2, Block-K, Plot-11, Road-17, Banani, Dhaka-1213

- Designed and implemented Layer 2 networking features in Embedded C for ARM-based carrier Ethernet switches running embedded Linux and VxWorks, working primarily in a command-line, no-IDE workflow with GCC and GDB.
- Studied, interpreted, and implemented networking standards and RFCs to ensure product compliance with international standards.
- Diagnosed and resolved system issues using performance analysis and kernel tracing tools (Wireshark, trace-cmd, KernelShark, ftrace), continuously improving stability and convergence.
- Collaborated with cross-functional teams in an agile environment, communicating in English with the overseas R&D team.

EDUCATION

Bachelor of Science in Computer Science & Engineering
Rajshahi University of Engineering & Technology, Bangladesh 2017 – 2022
CGPA: 3.11/4.0

TECHNICAL SKILLS

- **Embedded Systems and Operating Systems:** VxWorks, Embedded Linux, FreeRTOS, Linux.
- **Microcontrollers:** STM32, ARM; processor low-power modes.
- **Peripherals / Serial Protocols:** GPIO, SPI, I2C, UART, Ethernet.
- **Networking:** TCP, UDP, TFTP; Layer 2 - VLAN, STP / RSTP / MSTP, LLDP, ERPS (G.8032), EAPS, SNMP, CFM/OAM.

- **Messaging Protocols:** MQTT, HTTP/HTTPS.
- **Programming Languages:** Embedded C, C++, PHP.
- **Web Design:** CSS, HTML, Bootstrap, jQuery, JavaScript.
- **Debugging & Tools:** GDB, Wireshark, trace-cmd, KernelShark, ftrace, Scapy.
- **Version Control:** Git, SVN.
- **Database:** MySQL.

ONLINE ACCOUNTS

- Codeforces: Yamin1999 (Solved 100+ Problems)
- Leetcode: Yamin1999 (Solved 70+ Problems)

PROJECT EXPERIENCE

Embedded C & Low-Level Development

- Developed and maintained C/Embedded C software for ARM-based Carrier Ethernet switches running embedded Linux.
- Debugged low-level software issues involving pointers, packed structures, memory management, byte-order conversion, and 64-bit counter handling.
- Performed kernel tracing, protocol analysis, and performance optimization using GDB, Wireshark, trace-cmd, and command-line development tools.

Layer 2 Protocol Development

- Designed, implemented, and enhanced Layer 2 networking features for Carrier Ethernet switches.
- Developed and maintained LLDP, ERPS (G.8032), EAPS, CFM/OAM, VLAN, MSTP, and SNMP features following IEEE and ITU-T standards.
- Collaborated with cross-functional teams to debug protocol behavior, resolve interoperability issues, and deliver customer-specific networking features.

TFTP Server Module (GitHub)

- Developed a standalone TFTP server implementing RFC 1350 for secure file upload and download between network devices and management systems.
- Implemented configurable server parameters including UDP port, timeout, retransmission retry count, and enable/disable controls through the CLI.
- Added support for RFC 2348 (Blocksize Option) and RFC 7440 (Windowsize Option) to improve transfer performance and protocol compliance.

THESIS

Title: Deep Convolutional Neural Network for Malaria Parasite Detection

Our thesis suggests an improved machine learning framework using a deep convolutional neural network (CNN). Using five-fold cross-validation, based on 27,578 single-cell images, the common accuracy of our new 13-layer CNN version is 96.41% which is close to any transfer learning method. we have also shown that our deep CNN model is simpler and more computationally efficient than the other existing transfer learning methods.

REFERENCES

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